Lifting Tool for 3500 Engines Series Cylinder Liners

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

Editor's Note: The use of this installation tool and the processes discussed within this Best Practice is not a substitute for applicable laws, standards and regulations and does not alter or limit the obligation of individuals who utilize the tool to fully comply with federal, state and local law and prudent safety measures relating to the use of lifting devices.

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1.0 Introduction

The 3500 series engine has 16 cylinders, therefore the same number of liners with an approximate weight of 18.3 kg each. Which makes the installation process (lifting and handling) a difficult task to be achieved because it can cause back injuries, bruises, bumps and extreme fatigue for the technicians (see Fig. 1).

Fig. 1. Liners handling and lifting process thru the installation process.

This Best Practice aims to facilitate the development of this task with the design of a lifting tool which is of easy, fast and safe use.

2.0 Best Practice Description

The tool applies to 3500 series engines liners. It consists of two simple straight top links, two more "L" shaped lower links both made of steel plate A514 g A 20mm x 4mm; also, two arcs in TIVAR® 1000 (attached data sheet) 20mm thickness and internal diameter equal to the outer diameter of the liner. All this goes together symmetrically by hexagonal bolts 3/16" x 1" 8 grade with their spacing washers, lock nuts and at the upper end it´s placed a ring to engage the lifting equipment, forming a clamp as a whole (as shown in the figure). The clamp closure which is secured with a 1/8" steel pin blocking the rings from closing prevents the rings from splitting.

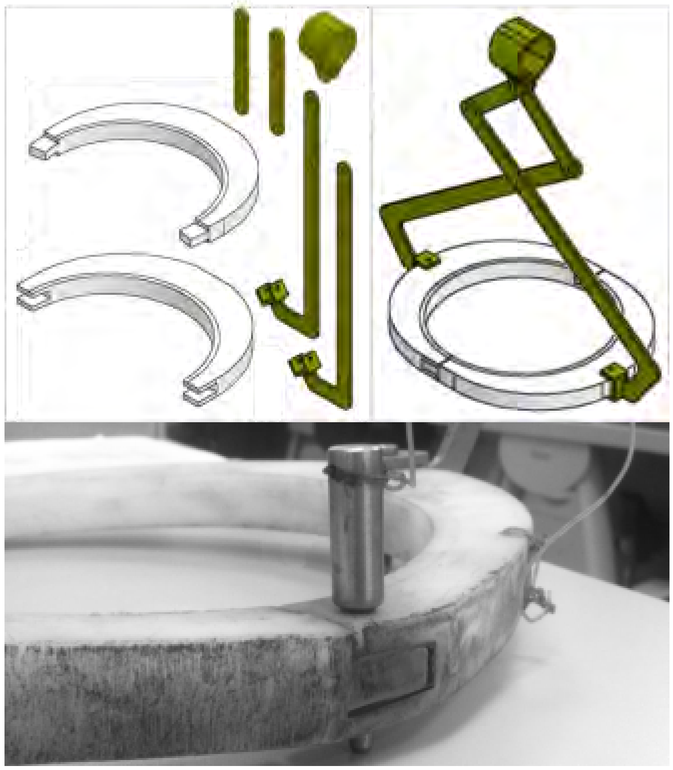

Fig. 2. Disassembly and assembly tool

The approximate weight of the tool is 1.7kg with a safety factor of 8 times the weight of the liner (18.3 kg). The tool is used for lifting from the floor level, where the baskets having the liners kit (16 in total) are, to the counter where the protective wrapping plastic is removed (the liner is prepared for installation). Then it´s raised back into the engine. The upper ring allows the required rotation to concentrically fit the liner along with the block hole where it will finally be installed.
NOTE: It is important that the lifting equipment to be used for this be of slow and precise movement, because when it comes to inserting the liner on the block; the hitting caused by the equipment without this feature should be avoided.

3.0 Implementation Steps

1. Check the status of the tool for physical defects (cracks, stretching, bending parts, etc.) before using it to ensure proper operation. Remove the lock pin to open the rings of the tool and hold the liner at the top edge; by closing and fastening the rings with the pin back again.
2. Lift the liner partially and slowly and make sure the tool closes automatically thru its scissor mechanism.
3. Fully lift to bring it to the counter where the plastic wrap is removed and the liner is prepared for installation. Hold the liner the same way and raise it to the engine block.
4. Close to the engine, the degree of freedom of the upper ring of the tool is used to turn the liner into the insertion angle of the block hole. Once the liner is secure enough, the locking pin is removed and the rings of the tool are open and the liner is finally released (Fig. 3 - 4a). An even safer and faster method is to rotate the roll over holding the engine to give such angle that the surface of the block holes is horizontal and perpendicularly receive the liner that is being hoisted (4b).

Fig. Operating Procedure.

4.0 Benefits

- Easy fabrication
- Easy operation
- It decreases the risks of extreme fatigue, back pains and bruises due to falls of the technicians.
- Lightweight for easy transport (with maximum weight of 1.7kg)

5.0 Resources Required

Drawings and date sheet of the tool (see attached)

6.0 Supporting Attachments / References

Quadrant EPP TIVAR® 1000 UHMW-PE, Virgin (ASTM Product Data Sheet)

Categories: Polymer; Thermoplastic; Polyethylene (PE); UHMW-PE

Material Notes:

- Meets FDA and USDA guidelines; 3-A Dairy-approved (natural)
- Reduces noise
- Self-lubricating
- Chemical-, corrosion- and wear-resistant
- No moisture absorption
- Non-toxic, low-friction surface
- Meets ASTM-D-4020-05 of 3.1 to 6.2-million molecular weight

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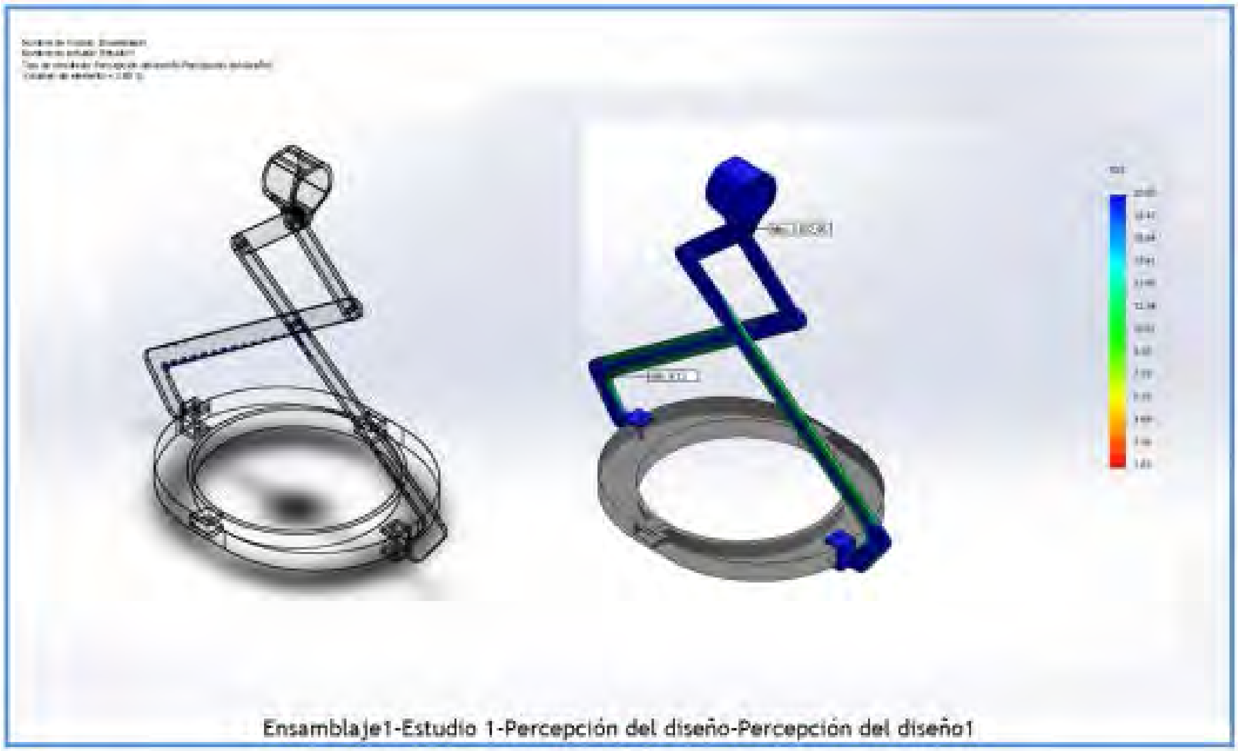
Physical Properties	Metric	English	Comments
Specific Gravity	0.930 g/cc	0.930 g/cc	ASTM D792
Water Absorption	<= 0.010 %	<= 0.010 %	immersion, 24hr; ASTM D570(2)
Water Absorption at Saturation	<= 0.010 %	<= 0.010 %	Immersion; ASTM D570(2)

Mechanical Properties	Metric	English	Comments
Hardness, Shore D	66	66	ASTM D2240
Tensile Strength	40.0 MPa	5800 psi	ASTM D638
Tensile Strength at 65°C (150°F)	2.76 MPa	400 psi	ASTM D638
Elongation at Break	300 %	300 %	ASTM D638
Tensile Modulus	0.552 GPa	80.0 ksi	ASTM D638
Flexural Strength	24.1 MPa	3500 psi	ASTM D790
Flexural Modulus	0.600 GPa	87.0 ksi	ASTM D790
Compressive Strength	20.7 MPa	3000 psi	10% Def; ASTM D695
Compressive Modulus	0.552 GPa	80.0 ksi	ASTM D695
Shear Strength	33.1 MPa	4800 psi	ASTM D732
Izod Impact, Notched	NB	NB	ASTM D256 Type A
Coefficient of Friction, Dynamic	0.12	0.12	Dry vs. Steel; QTM55007
Sand Slurry	10	10	1018 Steel = 100
Limiting Pressure Velocity	0.105 MPa-m/sec	3000 psi-ft/min	4:1 safety factor; QTM 55007

Fig. 4. Data sheet TIVAR® 1000



Nombre	Tipo	Mín.	Máx.
Factor de seguridad1	Automático	8.13223 Nodo: 6471	5812.06 Nodo: 7848

Fig. 5. FEA Analysis, Strength Calculations, Design Factor.

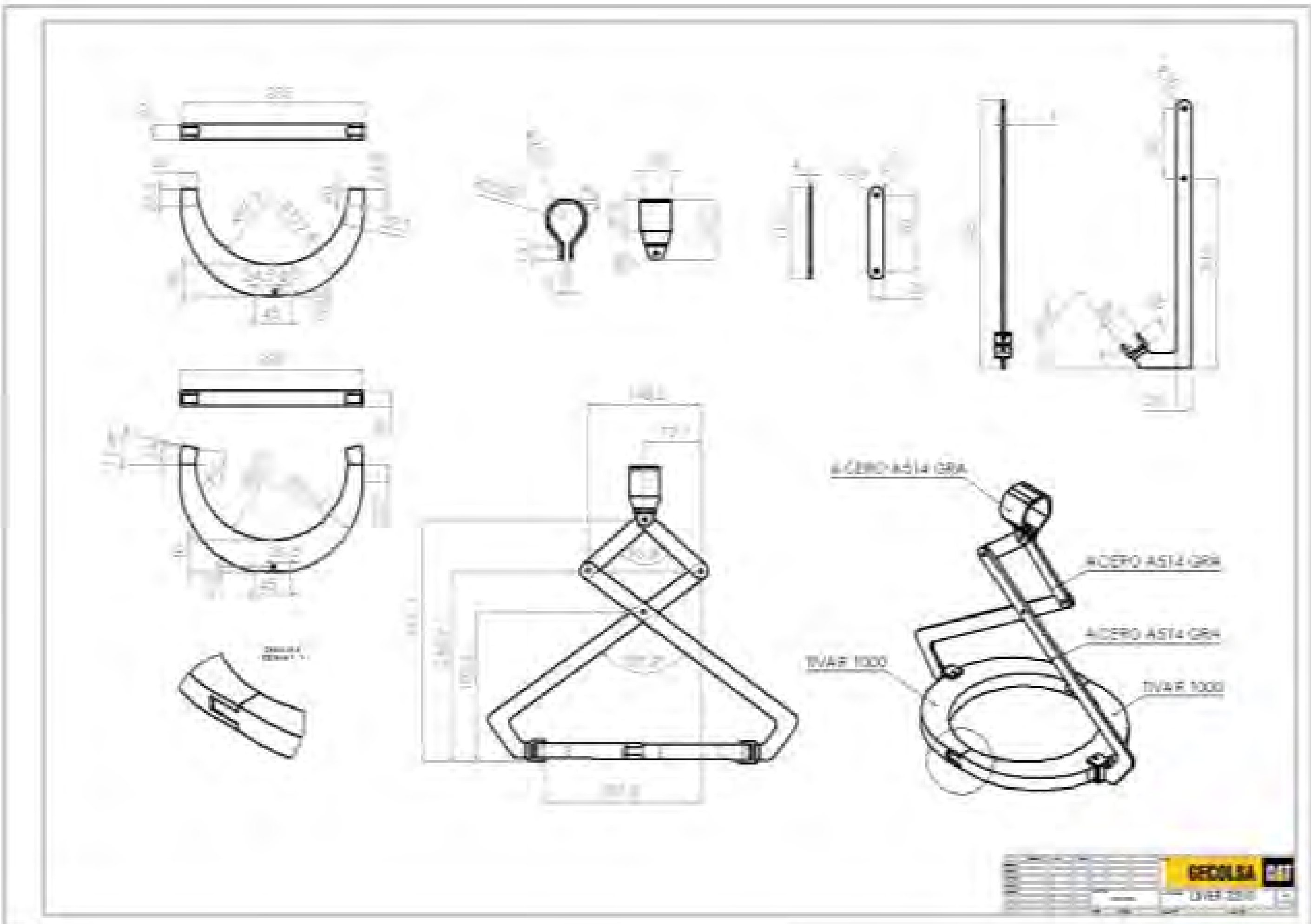


Fig. 6. Drawing of the tool

7.0 Related Best Practices

None

8.0 Acknowledgements

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